

# The Ghost Autonomy Technology Strategy

From provable boundary to accountable autonomy:  
the program of record, 2026–2028

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Status date | July 16, 2026                                                                   |
| Horizon     | Thirty months, gated; supersedes the EPU Master Roadmap of Feb. 2026            |
| Audience    | The program itself; investors and prospective partners read the same text       |
| Companion   | <i>Navigating by Difference</i> (Vol. II) — the comparative guardrail framework |

## THE TWO VOLUMES, AND THE CLAIM LADDER

This volume states what Ghost builds, proves, sells, and refuses to attempt, on what schedule and under which kill criteria. Volume II derives, from the five operating programs pursuing full autonomy, the fifteen guardrails (G1–G15) that bound every decision here; they are cited inline as **[Gn, Vol. II]** and function as standing constraints — a proposal that violates one is rejected without further argument. Both volumes carry the program’s evidence ladder: **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[DEMONSTRATED]** running software; **[PROJECTED]** an engineering estimate awaiting hardware; **[NOTIONAL]** a research direction, quotable as intent only.

Grounded in the program corpus: Ten Articles (May 2026); EPU–Ghost Autonomy Documentation v3.0 (June 2026); Provably Bounded compendium; Temporal State Management; program coverage map (Feb. 2026, rev.); CEO Research Opportunity Assessment (July 9, 2026); Governed Knowledge Systems trilogy (July 2026). Internal sources are listed in Appendix A.

# Contents

---

|                                         |    |
|-----------------------------------------|----|
| Executive Summary                       | 2  |
| 1 Strategic Intent                      | 3  |
| 2 The Thesis and Its Standing           | 3  |
| 3 Portfolio and Architecture Strategy   | 4  |
| 4 The Proof and Validation Program      | 5  |
| 5 Certification and Regulatory Strategy | 6  |
| 6 Commercialization Strategy            | 7  |
| 7 Intellectual Property and Publication | 8  |
| 8 Organization and Capital              | 8  |
| 9 Program Governance                    | 9  |
| 10 Contingency Branches                 | 9  |
| 11 Milestones and Measures              | 10 |
| 12 How to Use the Two Volumes           | 10 |
| A Internal Sources of Record            | 11 |

## Executive Summary

---

Ghost Autonomy's strategy is to become the **enforcement-and-accountability layer for bounded autonomy**: a physically fail-closed actuation boundary, an evidence-admissibility discipline, and a constitutionally governed memory — delivered first as safety-case *instrumentation*, then as a component-grade *guard ECU*, and only afterward, behind an explicit and reversible gate, as custom *silicon*. Ghost does not operate fleets, does not build vehicles, does not take sides in the sensor war, and does not use the phrase “Level 5” as a destination [G3, Vol. II][G7, Vol. II][G15, Vol. II][G13, Vol. II].

The strategy rests on three load-bearing claims, each carried at its honest standing. C1: that command authority can be made *risk-monotone* and enforced below software, terminating in a non-programmable analog veto — partially proven (the box-case algebra), witnessed once [MEASURED] (the ~32 ns FPGA veto), and open at exactly two points the program has already named: the one-sided trusted-scalar bound (P-1) and the liveness frontier (P-2). C2: that verification placed *before* fusion, plus a conjunctive conformance gate over six acceptance criteria, converts part of the industry's unbounded validation problem into bounded property verification — proposed [PROPOSED], and the basis of the first product. C3: that trustworthy autonomy requires *constitutional memory* — append-only canon, anti-silent-drift, priced recall — specified in the July 2026 trilogy and carried as notional [NOTIONAL] until a validation program runs.

The thirty-month program of record proceeds in three stages with falsifiable exits. Stage  $\alpha$  (through Q1 2027) closes P-1 and P-2, ships the Conformance Instrument against a public automotive dataset, and opens one assessor conversation; its failure modes are public re-scoping, not quiet death. Stage  $\beta$  (2027 through mid-2028) embodies the boundary in an FPGA guard ECU, converts the program's one measured number into a measured *family* under fault injection and automotive vibration profiles, and instruments one partner track pilot as a UL 4600-style evidence pack. Stage  $\gamma$  (gate at Q3 2028) decides silicon on economics rather than romance: if commodity safety hardware plus Ghost's architecture meets the contract, the EPU remains a research option and the company remains a layer supplier — an outcome this document declares legitimate in advance.

Everything else in this volume is the machinery that makes those commitments enforceable: a portfolio decision that concentrates capital on the boundary wedge and formally retires the decorative-mathematics register; a certification strategy that formats every artifact as a safety-case exhibit from birth; a commercialization ladder priced for the sub-\$30,000-vehicle cost curve [G4, Vol. II]; an IP posture of *publish the proofs, patent the mechanisms, keep the implementations*; an organization plan that converts a single-researcher program into an externally validated one; and a governance chapter that makes the claim ladder, the guardrails, and named kill criteria binding on the program itself — including on its founder.

# 1 Strategic Intent

## 1.1 Mission and identity

The mission, in one sentence: *make autonomous systems that can refuse, remember, and be audited — and make those three properties purchasable by the programs that drive.* The identity that follows, fixed by the synthesis of Volume II: Ghost is a **layer supplier** to the autonomy industry, occupying the two strata the operating field has left vacant — enforcement below the programmable stack, and governed memory above it. The claim is deliberately narrow. Perception, prediction, planning, mapping, teleoperation, ride operations: all are explicitly *not* Ghost’s business, and every one of them is somebody’s \$10B lesson [G3, Vol. II].

## 1.2 Objectives to end-2028

Four objectives, each with a measurable outcome and no others:

- **O1 — Prove the boundary.** P-1 and P-2 closed and submitted to peer review; the measured-quantity count raised from one (the ~32 ns veto) to at least six hardware-witnessed figures on the Stage- $\beta$  ECU.
- **O2 — Make the safety case checkable.** The Conformance Instrument adopted as working tooling in at least one external program’s safety-case preparation, with Ghost artifacts appearing verbatim in a filing or audit binder [G8, Vol. II].
- **O3 — Earn the assessor moat.** A standing technical dialogue with at least one accredited assessor (TÜV/UL/exida stratum), evidenced by review letters or a pre-assessment engagement [G12, Vol. II].
- **O4 — Keep every public claim at its evidenced standing.** Zero claim-ladder violations across all program surfaces, audited quarterly; the standing of a claim may move *down* in public at any time, and that movement is reported as method, never concealed [G13, Vol. II].

## 1.3 Scope boundaries (the “never” list)

For the horizon of this document Ghost will not: operate a passenger service or own a fleet [G3, Vol. II]; design or manufacture a vehicle [G7, Vol. II]; advocate any sensor modality [G15, Vol. II]; publish an undated capability promise or an SAE level as destiny [G1, Vol. II][G13, Vol. II]; present volume — of miles, pages, or parameters — as evidence [G14, Vol. II]; or tape out silicon before the Stage- $\gamma$  gate criteria are met in writing. Scope boundaries are governance objects: changing one requires the same public, logged, challengeable event the architecture demands of any change to operating truth.

# 2 The Thesis and Its Standing

The program’s three claims are stated here the way its own corpus requires — with their evidence, their open points, and what would break them.

## 2.1 C1 — Contraction can be enforced below software

**Claim.** As warrant for a benign reading of the scene falls, the set of admissible commands may only contract; the terminal contraction is executed by non-programmable hardware — comparators and latches with no rewritable state in the guard path — so that no fault, exploit, or model error above it can widen authority. **Standing.** The actuation algebra is proven for convex box envelopes (projection = coordinatewise clip; firm non-expansion; feasible-case exactness of the two-stage clamp) [PROVEN]. The veto exists as an FPGA witness at ~32 ns intervention latency [MEASURED] — the only measured number in the program, a fact this strategy treats as its most honest asset. **Open.** P-1: the one-sided error bound on the trusted scalar  $\varphi$  (the

corpus's own ledger grades it CRITICAL: "without it the contract is aspirational"). P-2: the liveness frontier — a gate that always reports maximum risk is vacuously safe and parks the vehicle forever. P-3: exactness beyond boxes (friction-circle geometry). **Breaks if.** P-1 is unprovable in any useful generality, or the liveness trials show the boundary cannot be simultaneously safe and useful. Both failures are visible, cheap, and scheduled — Stage  $\alpha$  exists to force them into the open.

## 2.2 C2 — Verification before fusion changes the validation economics

**Claim.** If evidence is admitted to the world model only after integrity, alignment, scope, and lineage gates (sensors as witnesses; correlated witnesses counted once; inadmissible testimony excluded, never averaged), and if release is governed by a *conjunctive* gate over the six acceptance criteria G1–G6, then part of the industry's unbounded scenario-coverage burden becomes bounded property verification — a third route between Waymo's saturation and Mobileye's channel-independence argument (Vol. II, Reading 2). **Standing.** Proposed [PROPOSED]; the gate semantics and criteria are fully specified in the corpus; nothing has been exercised against automotive data. **Open.** The transfer obligation: the corpus's empirical anchor is gasification, not driving — its own text marks automotive transfer as an open validation obligation. **Breaks if.** The admissibility layer, run against real automotive perception streams, either rejects so much evidence that liveness fails or admits so much that the structural claim adds nothing beyond existing SOTIF practice. Stage  $\alpha$ 's dataset campaign is designed to measure exactly this.

## 2.3 C3 — Accountability requires constitutional memory

**Claim.** Autonomy at fleet scale fails on governance before it fails on capability: silent drift, unexplainable decisions, unaudited updates, correlated platform failure. The remedy is memory that is constitutive rather than auxiliary — append-only canon with challengeable amendments (anti-silent-drift), verification burden that *increases* with residence time, recall as a priced contract, and quarantine between missions. **Standing.** Notional [NOTIONAL] by the trilogy's own declaration; the doctrine's automotive instantiation (between-drives quarantine, signed update ledgers) is proposed [PROPOSED]. **Why it stays in the strategy at all:** the regulatory settlement is converging on it — China's DSSAD data-recorder mandate (July 2027), ISO/PAS 8800's runtime monitoring and data governance, California's structured safety case — and no operating program offers a governed answer (Vol. II, Reading 3). **Breaks if.** No design partner values governed fleet memory enough to pilot it by the Stage- $\gamma$  gate; in that branch the trilogy remains published specification and the company loses nothing but an option.

## 2.4 The substrate, demoted to an option

The EPU — variational cores, event registers, the PICAPD ISA, elliptic-integral primitives in hardware — is deliberately *not* one of the three claims. Every silicon figure in the corpus ( $\sim 3.4$  ns decision path,  $10^6$  decisions/s,  $\sim 23\times$  GPU power efficiency) is projected [PROJECTED], and the corpus's provenance ledger already forbids those figures from masquerading as built. The strategy therefore treats custom silicon as a *real option* the earlier stages may or may not exercise: Stage  $\beta$ 's guard ECU deliberately uses commodity parts (lockstep safety MCU, FPGA, analog path) so that the Stage- $\gamma$  decision is an economics question — does the boundary's market require the EPU? — rather than an identity question. The founder's mathematics loses nothing in this demotion; the AGM primitive [PROVEN] and the reduced-order physics kernel remain the program's research spine and the candidate trust anchor for  $\varphi$ . But if commodity hardware meets the contract, Ghost remains a layer supplier and the EPU remains a beautifully documented research program. That sentence is in this document so it can be quoted back to the program later.

# 3 Portfolio and Architecture Strategy

### 3.1 The four layers and their build/adopt boundaries

**Layer 1 — The Conformance Instrument (software; Stage  $\alpha$  product).** The G1–G6 conjunctive gate, the claim ledger (established/proposed/notional standings as machine-checkable metadata), and the twin-world replay harness, packaged as tooling a safety team can run against its own stack’s traces. Build: the gate semantics, ledger schema, replay format — these are the differentiation. Adopt: scenario formats and dataset tooling from the existing SOTIF/scenario-based-testing ecosystem; nothing is gained reinventing OpenSCENARIO.

**Layer 2 — The Guard ECU (hardware; Stage  $\beta$  product).** A stateless enforcement node — lockstep safety MCU plus FPGA envelope logic plus the analog veto path — with direct drive-by-wire authority, timestamped lineage-carrying sensor taps, and safe-side numerics for every gating scalar. Build: the risk-banded envelope logic, the veto path, the witness log. Adopt: qualified automotive compute (the ASIL-D MCU ecosystem exists precisely so that new entrants need not fabricate trust), ISO 16750-qualified packaging practice, existing drive-by-wire interfaces. The partner’s perception and planning stack survives untouched as the fallible proposer — integrability is the product [G7, Vol. II].

**Layer 3 — The EPU substrate (silicon; Stage  $\gamma$  option).** Proceeds only on the gate of §6.3. Until then the ISA specification continues as publishable research, and every figure stays [PROJECTED].

**Layer 4 — The Memory Module (software/governance; subordinate wedge).** The trilogy’s fleet-facing instantiation: signed update ledgers, between-drives quarantine, priced-recall budgeting for fleet learning pipelines, formatted against ISO/PAS 8800’s monitoring and data-governance clauses. It is sold, when it is sold, *through* the credibility of Layers 1–2, to the fleet-learning problem every operator now has and none has governed — including the correlated-failure semantics (fleet-level minimal-risk condition) that Wuhan demonstrated and no one has claimed [G5, Vol. II].

### 3.2 Portfolio concentration and formal retirements

Strategy is subtraction. In line with the program’s July 2026 opportunity assessment, this document makes the following dispositions binding. **Concentrated:** the actuation boundary (Layers 1–2) receives the capital, the calendar, and the founder’s hours. **Supporting:** the physics kernel (AGM/elliptic reduced-order models) continues as the  $\varphi$  trust-anchor candidate, with its automotive transfer treated as an experiment, not a premise; if transfer fails,  $\varphi$  falls back to classical envelope estimation and *the boundary still ships* — C1 does not depend on the kernel. **Subordinate:** the Memory Module as Layer 4. **Maintained, uncanceled:** adjacent instruments of the research program (e.g., the dynamod composition tool) as internal capability, not product. **Retired:** the decorative-mathematics register — the cosmological-constant coincidence, the large-cardinal framings — already ordered out of safety paths by the corpus’s own review; this strategy extends the retirement to all external surfaces [G13, Vol. II]. **Superseded:** the February 2026 “EPU Master Roadmap” (36-month tape-out schedule, named OEM launch partners, pre-discipline register). Its ambition survives; its sequencing, its silicon-first premise, and its rhetoric do not. This volume is the roadmap of record.

## 4 The Proof and Validation Program

### 4.1 The proof ledger as the R&D plan

The corpus’s Ledger B is adopted here, unmodified, as the research agenda — the program’s honesty made schedulable. Sequenced: **P-1** one-sided  $\varphi$  bound (CRITICAL; Stage  $\alpha$ , first); **P-2** liveness frontier characterization (Stage  $\alpha$ , concurrent — safety without usefulness is parking); **P-3** non-box actuator geometry (Stage  $\beta$  entry criterion for any partner whose actuation set is not a box, i.e., all of them eventually); **P-4** null-band calibration under real vibration and weather (Stage  $\beta$ , tied to the qualification campaign); **P-5** clock/synchronization semantics at multi-agent scale and **P-6** write-ahead-log workload

profiling (Layer 4 debts; scheduled only when the Memory Module earns a pilot); **P-7** certification-boundary marking (runs continuously — every artifact labels what is inside the certified perimeter and what is not). Combined estimate, per the ledger: 90–131 person-days of focused theoretical work — small enough to be a seed-stage deliverable, decisive enough to be the thesis.

## 4.2 Stage $\alpha$ — Instrument (now through Q1 2027)

Deliverables: P-1 and P-2 closed and submitted; Conformance Instrument v0 released and exercised against a public automotive perception dataset, with the admissibility layer’s accept/reject behavior published as measured distributions (the direct test of C2’s transfer risk); one assessor dialogue opened with the artifacts as exhibits; the claim-ladder audit institutionalized. Falsifiers, stated in advance: P-1 unprovable as stated  $\Rightarrow$  public re-scoping of the  $\varphi$  contract; liveness empirically vacuous  $\Rightarrow$  the structural thesis has failed and §10’s contingency branch executes. Cost profile: proofs and software only [G3, Vol. II].

## 4.3 Stage $\beta$ — Boundary (Q1 2027 through mid-2028)

Deliverables: the guard ECU on commodity parts in a hardware-in-the-loop rig; fault-injection campaigns; the measured family (intervention-latency distribution, missed-intervention bound, envelope-bleed profile, veto-path timing under ISO 16750-3 vibration — closing the coverage map’s own GAP cluster, budgeted at 55–80 person-days); one instrumented retrofit pilot on a partner test track, packaged as a UL 4600-style claims-argument-evidence binder [G8, Vol. II][G9, Vol. II]; P-3 and P-4 closed. Falsifiers: veto timing fails under vibration  $\Rightarrow$  qualification before any scale claim; no partner hosts a pilot on the Stage- $\alpha$  evidence  $\Rightarrow$  the constraint is market, not mathematics, and §10 branches.

## 4.4 External validation as a design requirement

The program’s known weakness is that its evidence base has one author. Stage  $\alpha$  therefore builds the correction into the machinery: peer-review submission of P-1/P-2; public release of the instrument for refutation (the corpus’s “stronger evidence beats stored consensus” applied to itself); an advisory bench spanning formal methods, functional safety, and automotive E/E architecture; university partnership for the HIL laboratory — Atlantic-Canadian engineering faculties are a practical, low-cost fit for the Halifax-based program. From Stage  $\alpha$  onward, no claim above [PROPOSED] enters an external document on the founder’s signature alone.

# 5 Certification and Regulatory Strategy

## 5.1 Standards mapping

The strategy formats Ghost’s native artifacts as the exhibits the regimes already ask for — arriving by construction where the field is being pushed by remediation:



| Regime                                                          | Ghost artifact that answers it                                                                                                                                                               |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO 26262 (ASIL)                                                | The guard ECU as a safety element out of context; lockstep MCU heritage; the analog veto as the highest-integrity path. Alignment pursued, certification never claimed early [G13, Vol. II]. |
| ISO 21448 (SOTIF)                                               | The conjunctive G1–G6 gate reframing unknown-unsafe coverage as consequence bounding; admissibility-layer measurements as SOTIF evidence of triggering-condition handling.                   |
| UL 4600                                                         | The claim ledger is a claims-argument-evidence case in machine-readable form; replay traces as evidence items; P-7 marks the certified perimeter.                                            |
| ISO/PAS 8800                                                    | Layer 4's quarantine, drift events, and priced-recall logs as the required AI runtime monitoring and data governance.                                                                        |
| IEEE 2846                                                       | The $\varphi$ contract and admissibility rules published as an explicit assumption registry — engaging, not ignoring, the RSS-seeded standard [G10, Vol. II].                                |
| Part 555 / AVEP; California structured case; China DSSAD (2027) | The evidence binder of Stage $\beta$ formatted for exemption petitions and structured-case filings; witness logs as native black-box compliance.                                             |

## 5.2 Jurisdictional sequencing

Develop and validate from Canada (neutral ground, university partnerships, no permit dependency for bench and track work); first regulatory engagements in the United States, where the safety-case turn is codified and the exemption grammar is public; treat China's 2027 data-recorder mandate as a market signal for Layer 4 rather than a near-term entry, and Europe's UNECE small-series routes as the channel that arrives with a Tier-1 partner, not alone. Ghost files nothing itself in this horizon; it makes its partners' filings stronger — the layer supplier's posture, everywhere [G8, Vol. II][G12, Vol. II].

# 6 Commercialization Strategy

## 6.1 The rung ladder and pricing philosophy

Revenue at every stage of belief [G11, Vol. II]: **Rung 1** (from Stage  $\alpha$ ): the Conformance Instrument licensed per program/seat to OEM and operator safety organizations and to assessors — priced as engineering tooling, sold on validation-cost reduction, valuable even to buyers who never adopt Ghost hardware. **Rung 2** (from Stage  $\beta$ ): the guard ECU as a component-grade unit — its bill-of-materials ceiling set by the sub-\$30,000-vehicle cost curve, i.e., a low-single-digit percentage of vehicle cost, or it does not ship [G4, Vol. II]. **Rung 3** (post- $\gamma$ , conditional): the substrate, licensed or supplied, only if the gate opens. **Rung 4** (parallel, opportunistic): Layer 4 fleet-memory governance subscriptions, entering through Rung-1 relationships. The instrument's economics deliberately mirror what Mobileye proved: the low rung funds credibility for the high one, and every customer teaches the program where the boundary binds.

## 6.2 Customers, channel, and the marketing that is not marketing

Named archetypes, from Volume II §7.3: mid-tier operators and OEM autonomy programs facing the 2026 safety-case regimes without leader-scale assurance headcount; Tier-1 suppliers seeking differentiated safety content (the channel through which components actually reach vehicles [G12, Vol. II]); the assessor stratum, for whom the instrument is tooling and Ghost artifacts are reference implementations; regulators, reached through partners' filings. The leaders themselves — Waymo-class integrators — are benchmarks and eventual acquirers of ideas, not near-term buyers; the strategy wastes no calendar courting them. Marketing is the published corpus, the refutation invitations, and the measured families: for a program whose product is accountability, epistemic conduct is the demand-generation engine [G2, Vol. II][G6, Vol. II].



Deep documentation remains NDA-gated exactly as the site practices; nothing public ever exceeds its standing.

### 6.3 The name decision

Before any external raise, the brand-collision hazard with the defunct Ghost Autonomy Inc. (~\$220M, shut down April 2024, OpenAI-backed, remembered for claims outrunning embodiment) is resolved deliberately: trademark and domain search, then either a distinct legal name with “Ghost” retained as program branding, or a clean rename. The data room carries a one-paragraph disambiguation that converts the collision into contrast: *they promised driving and shipped none; this program promises only what it can grade, and grades everything*. Deadline: with the Stage- $\alpha$  raise, whichever comes first.

## 7 Intellectual Property and Publication

Three tracks, resolving the corpus’s current “Patent Pending & Trade Secret” posture into policy. **Patent** the mechanisms where infringement is detectable in products: the risk-banded envelope enforcement path, the analog-veto integration pattern, the admissibility-gated fusion interface, the residence-time-indexed verification scheme of the memory doctrine. **Publish** the theory and the specifications — P-1/P-2 proofs, the G1–G6 gate semantics, the  $\varphi$  contract, the ISA as research — because publication is simultaneously defensive prior art, the assessor-facing credibility engine [G2, Vol. II], and consistent with the house rule that claims live in public at their true standing. **Retain as trade secret** the implementations: RTL, calibration procedures, null-band parameters, the instrument’s internals. The boundary between tracks is P-7’s certified-perimeter marking, reviewed at each stage gate. One standing rule: no patent filing may assert, even in boilerplate, a capability the claim ladder has not certified — the IP portfolio inherits the epistemics [G13, Vol. II][G14, Vol. II].

## 8 Organization and Capital

### 8.1 From one researcher to a validated program

The founder remains the architect and the theory lead. The first three hires, in order, exist to make the evidence base plural: a **formal-verification engineer** (owns P-1/P-3 mechanization and the instrument’s proof kernel); a **functional-safety architect** with assessor-side experience (owns the standards mapping, the evidence binders, and the assessor dialogue — the moat-builder [G12, Vol. II]); a **mixed-signal/FPGA engineer** (owns the veto path, the HIL rig, and the measured family). The advisory bench (formal methods academic; retired assessor; automotive E/E architect) signs reviews rather than lending names. The university HIL partnership doubles as recruiting pipeline and as the independence mechanism of §4.4.

### 8.2 What each stage’s capital buys

Stage  $\alpha$  is a proofs-and-software budget: three salaries’ equivalent, dataset compute, and travel to two assessor/standards venues — deliberately small enough that its falsifiers are affordable. Stage  $\beta$  adds the bench: qualified MCU/FPGA hardware, the HIL rig (cost-shared with the university), the vibration/qualification campaign (the corpus’s priced 55–80 person-days), and the pilot’s instrumentation. Stage  $\gamma$ ’s silicon budget is not planned here — by design, it is the gate’s output, sized only if the gate opens, and raised against Rung-1/2 revenue evidence rather than projections [G14, Vol. II]. The program never raises against a fleet, a city count, or a ride metric; it raises against theorems closed, numbers measured, and binders shipped.

## 9 Program Governance

---

The architecture's constitutional principles apply to the company that builds it. **The claim ladder is the operating system:** every external sentence carries a standing; the quarterly audit (O4) is run against all surfaces — site, knowledge base, investor materials, patent filings — and downgrades are published, not smoothed. **The guardrails are standing law:** each quarter, every active initiative is tested against G1–G15; a violation is resolved by changing the initiative or, with a logged and reasoned amendment, the guardrail — never by silence (anti-silent-drift, applied to strategy). **Kill criteria are named at birth:** the memristive substrate bet already carries one in the corpus; this document adds them for the EPU (the §6.3 gate), the physics-kernel transfer (automotive floor met in Stage  $\alpha$  or fallback executes), the Memory Module (design partner by the  $\gamma$  gate or it remains specification), and the liveness thesis itself (§4.2). **The decision registry** — already the program's habit — records each gate outcome with its evidence, so that in 2028 the program can be audited against this very document. A strategy that demands accountability from autonomous systems and exempts itself would be the one inconsistency no investor should forgive.

## 10 Contingency Branches

---

Five branches, pre-committed. **(1) P-1 fails as stated:** the  $\varphi$  contract is re-scoped in public; the boundary ships with an empirically calibrated gate at a lower certification claim; C1 survives in weakened form and the instrument business is unaffected. **(2) Liveness proves vacuous:** the structural thesis is falsified; the program pivots to the instrument-only business (safety-case tooling has standalone demand) and says why — the honest exit, cheap because it was scheduled early. **(3) Kernel transfer fails:**  $\varphi$  falls back to classical estimators; the physics kernel returns to research; no product dependency breaks (§3.2's decoupling, exercised). **(4) No pilot partner after Stage  $\alpha$ :** the constraint is demand, not proof; the instrument extends to adjacent bounded-autonomy markets with SOTIF-shaped problems (industrial mobile robotics, off-road and mining autonomy) while automotive relationships mature — the layer thesis is domain-portable even though this strategy is automotive-first. **(5) Capital arrives thin:** the program runs the standards-and-licensing route — instrument licensing, assessor tooling, contributed reference work in standards bodies — extending runway without violating a single scope boundary. No branch involves fleets, vehicles, undated promises, or silence.

## 11 Milestones and Measures

| Checkpoint               | Committed outcome                                                                               | Measure of record                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Q4 2026                  | P-1 proof complete internally; instrument architecture frozen; claim-ladder audit #1 published  | Proof draft + audit report                                        |
| Q1 2027 ( $\alpha$ gate) | P-1/P-2 submitted; Instrument v0 public with dataset campaign results; assessor dialogue opened | Submissions; measured accept/reject distributions; meeting record |
| Q3 2027                  | Guard ECU alive in HIL; first fault-injection family measured                                   | $\geq 3$ new measured quantities                                  |
| Q1 2028                  | Vibration/qualification campaign closed (GAP cluster); P-3/P-4 closed                           | Qualification report; proofs                                      |
| Mid-2028 ( $\beta$ gate) | Track pilot instrumented; UL 4600-style binder delivered to partner and assessor                | The binder itself                                                 |
| Q3 2028 ( $\gamma$ gate) | Silicon decision taken on written criteria; Layer-4 pilot decision taken                        | Gate memorandum in the decision registry                          |

The standing dashboard, replacing the industry's ride counter: theorems closed ( $0 \rightarrow 2$  by  $\alpha$ , 4 by  $\beta$ ); measured quantities ( $1 \rightarrow \geq 6$ ); external validations ( $0 \rightarrow \geq 2$  peer-reviewed or assessor-reviewed); pilot integrations ( $0 \rightarrow 1$  HIL, 1 track); claim-ladder violations (0, always); and — the only vanity metric this program is permitted — the number of its claims other people have tried and failed to refute.

## 12 How to Use the Two Volumes

This volume decides; Volume II constrains. When a new initiative is proposed, it is drafted against this document's stages and scope boundaries, then tested against the fifteen guardrails; when the field shifts — a comparator collapses, a regime changes, a new pole appears — Volume II is revised first and this volume amended through the governance of §9. Investors should read them in the order they were written: the field, then the program. The program reads them in the opposite order every quarter: first what we said we would do, then what the world has since made of everyone who did otherwise.

## A Internal Sources of Record

---

Ten Articles on Provably Bounded Autonomous Driving, consolidated edition (May 2026, 89 pp.); EPU–Ghost Autonomy Complete Technical & Architecture Documentation v3.0 (June 2026, 94 pp.); Provably Bounded compendium (367 pp.; Part III carries provenance ledgers A/B); Temporal State Management monograph; Program Coverage Map (Feb. 2026; rev. 2 with Ledger B proof debts P-1–P-7); CEO Research Opportunity Assessment (Office of the CRO, July 9, 2026); Governed Knowledge Systems trilogy: Symbiotic Search Fabric white paper, Retrieval Covenant (Thesis II), Price of Recall (Thesis III) (July 2026); comprehensive co-design treaty (June 2026); PICAPD ISA Official Specification; EPU Demonstration Suite execution plan; superseded: EPU Master Roadmap v1.0 (Feb. 21, 2026). External sources for all field claims are carried in Volume II’s References (83 entries, dated through July 16, 2026); standards cited: SAE J3016; ISO 26262; ISO 21448; ISO/PAS 8800; UL 4600; IEEE 2846.